MachineVision-Based Fall Detection System using MediaPipe Pose with IoT Monitoring and Alarm

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***Abstract*— The incidence of falls is more commonly experienced by the ageing population globally, due to their increasing frailty. Fall detection systems are being developed combined with machine learning approaches that include those wearable devices, ambience-based systems, and vision-based systems. Wearable systems are omnipresent but intrusive while ambience-based systems are temperature-dependent that require a dedicated GPU. Challenges on vision-based fall detection systems include large computational cost, use of specialized camera, and system integration into smartphones. This research aims to propose a vision-based fall detection system ported on a smartphone application, that utilizes deep learning trained and tested from multiple RGB camera setups. In an event of a fall, the system is designed to provide an on-premises auditory alarm, an IoT notification, and a real-time video feed via the smartphone application. Using the pretrained MobileNetv2 CNN-based MediaPipe Pose and Random Forest Classifier for fall detection, experimental results show high-performance evaluation metrics.**

Keywords*— *CNN, Fall Detection, Internet of Things, Machine Vision, MediaPipe Pose

I. INTRODUCTION

Aging is inevitable and a natural change that happens to everyone all over the world. It has crucial effects on most of the systems including nervous, metabolic, sensory, cardiovascular, respiratory, and musculoskeletal [1]. The United Nations projected that by 2050, older adults will be 1 in 6 people in the world [2]. In the Philippines, it is projected that by 2030, 9,560,000 of the population will be 65 years and above [3]. Due to the increasing number of the aging population, the fall detection system among the elderly has piqued the interest of health professionals and the scientific community. Falls are ubiquitous and sometimes can be very costly among seniors. Every year almost 36 million older people are at risk of falling down and almost 3 million are treated in emergency departments because of a fall injury. According to the World Health Organization, 64% of falls can result in fractures, 44% of fear of falling or trauma, and 32% of hospital admissions. Falls can also result in "post-fall syndrome" that includes dependence 32%, loss of autonomy 14%, confusion 22%, immobilization 4%, depression 2%, and restrictions in daily life [4]. Efficient detection and classification of falls can be beneficial to achieve a quicker response to possible emergencies.

Fall detection systems for the elderly have been a trend nowadays due to the increasing aging population. It has been classified into three: wearables, ambience-based and vision-based. The substantial solutions to detect a fall activity have been deemed satisfactory, however, there are still gaps within each category. Studies that feature wearable devices [5]–[11], allow the users to wear it in different settings and still, detect fall activity. However, it is intrusive and is one user at a time only. Ambience-based systems like studies [12], [13] have garnered high accuracies when it comes to detection, however, they are temperature-dependent and desktop-based. Vision desktop-based systems, though excellent in monitoring like [14]–[22], are dependent on high computational learning algorithms that require a GPU. Some of the vision based-embedded systems did not use deep learning, some used specialized cameras, and some were relatively expensive [14], [23]–[27]. This is where the proposed study stood out since it has a monitoring capability, and incorporates an embedded system, that requires less computational power compatible with an RPi. Just like other vision-based studies for fall detection, [14]–[21], [23], [28] the proponents of this study prioritize monitoring over privacy since in scientific studies it is best to see the scenario, in reality, to make a recommended observation and arrive at a definitive conclusion.

This study proposes a non-invasive vision-based fall detection system that works on different camera setups using machine learning with IoT-based monitoring and alarm. The system alerts contact person(s) via auditory alarm and/or through IoT notification when a fall activity is detected. It also comes with visualization of object tracking and detection. A deep learning algorithm was used in classifying objects and detecting fall activity.

The paper is organized such that Section II gives details about related works on the proposed system. Section-III presents the proposed methodology and Section-IV discusses the experimental results and analysis. Finally, Section V concludes the paper and recommends future works.

II. Related Work

Fall detection activity has been a trend today as there has been significant aging in population happening in most parts of Asia and Europe. The Philippines may not be high in aging compared to its neighboring Asian countries like China, Singapore, and Japan [2]. But, Filipinos' value for extended family has played a huge role in developing this study. The popularity of fall detection amongst the elderly has opened developmental studies like wearable devices, ambience-based, and computer vision. The local studies available are supervised machine learning-based fall detection [5] and fall detection wearable devices interconnected through the ZigBee network [6] both used accelerometer devices to detect a fall. However, the accelerometer is an intrusive element. Other foreign studies that have utilized wearable devices are [9], [10], and [11]. These studies have intrusive elements that do not support multiple camera setups detection and do not feature an IoT monitoring and alert. The proposed study eliminates the wearable device as it is vision-based, capable of multiple camera arrangements detection, and has an IoT monitoring and alert.

Current ambience-based fall detection systems are [12] and [13]. The first study [12] uses a radar sensor and uses waves to accurately detect fall detection. It also uses a desktop computer, signal generator, and oscillator. The latter depends on the distribution of space occupied by a human. However, both of the studies have no IoT monitoring, desktop-based [12], [13] and used an Arduino microcontroller [13]. The proposed study is an IoT-based monitoring and alarm that is embedded in RPi. The RPi has better clock speed and is capable of giving remote access that is unique in every home network providing privacy for its users. It is also ideal for python programming that has wide libraries which can be used in deep learning and image processing.

Computer Vision or Vision-based systems can be further subdivided into two categories: desktop/laptop-based and embedded. Some studies that utilized computer vision that are laptop/desktop-based are [14], [15], and [16], [17], [18], [19], [20], [21], and [22]. Moreover, unlike the proposed study, studies [14]–[22] do not have an IoT monitoring and alert. Another disadvantage is that these studies [14]–[21] require a dedicated GPU for training and testing. The proposed study is embedded on RPi 4 microcontroller and not desktop-based to avoid heavy computational power. Researches including [23], [24], [25], [26], and [27] are the studies that are embedded vision-based fall detection systems. Nevertheless, studies [23]–[27] have no physical alarm and do not support IoT monitoring. Studies like [23], [24], [26], [27] considered fixed angle detection only while studies [23]–[26] have no IoT alarm or even a notification.

Table 1 presents the comparison of existing fall detection systems from different aspects. For vision-based fall detection systems, a comparison for type of embedded system used, camera, system size, cost, machine learning used, and accuracy is explained in Table 2. In summary, the proposed study is a non-wearable automated fall detection system for the elderly that is trained in multiple camera setups using a single RGB camera that uses a deep learning-based algorithm embedded on a microcontroller. The system includes a physical alarm and IoT-based monitoring and alert to notify contact person/s.

Table 1. Survey of Existing Fall Detection systems

	Wearable	**Ambience**	**Vision (Desktop)**	**Vision (Embedded)**	**Proposed**
Real-Time Visualization	N/A	Low	High	Medium-High	High
Actual Processing of Data	Desktop/ Server/ Embedded	Desktop/ Server	Desktop	Desktop/ Server/ Embedded	Embedded
Cost	Low-Medium	Low-High	High	Low-High	Low
Size	Small-Medium	Small-Medium	Large	Small-Large	Small
Sensors	Gyroscope, Accelerometer Smartphone, Smartwatch	Ultrasonic, Radar	RGB camera, Kinect, depth camera	RGB camera, Kinect, CMOS camera, depth camera	RGB camera

Table 2. Survey of Vision-Based Fall Detection Systems

Study	**Embedded System**	**Sensor/ camera**	**Size**	**Cost**	**Classifier**	**Model Accuracy**
B.S. Kumar (2019)	RPi 3B	RGB Camera	Small (Compact)	Low	No Deep Learning: OpenCV	90.65
Y. Ogawa and K. Naito (2020)	2 RPi 3B	IR Sensor	Small (Compact)	Medium	Voting	97.75%
T. Tsai and C. Hsu (2019)	NVIDIA Jetson Tx2	Kinect	Medium (Not Compact)	High	Thinning. DNN	99.2%
C. Menacho and J. Ordoñez (2020)	NVIDIA Jetson Tx2	Kinect	Large (Not Compact)	High	Modified CNN	88.55%
X. Yang, S. Tang, T. guo, K. Huang, and J. Xu (2020)	ZYNQ(ARM+FPGA).	CMOS camera	Medium (Not Compact)	High	CNN	92%

III. Methodology

Mobile communication devices combined with Internet of Things can provide improved remote access and control to systems. In this paper, a non-invasive automated fall detection system was developed and trained on multiple camera setups using deep learning with IoT-based monitoring and alarm. The system is designed to provide an on-premises auditory alarm, an IoT notification, and a real-time video feed via a mobile application.

A. Hardware & Mobile Application

Figure 1 shows the block diagram of the proposed system. The RPi camera v2 module, placed optimally on an elevated area, captures the input video. The programmed Raspberry Pi 4B then processes the video data to detect a fall activity. If a fall activity was detected, an on-premises sound alarm would be triggered, and a notification would be sent over a cloud data server thru the internet onto a mobile phone application. The IoT notification thru the mobile phone is composed of a sound and display message.

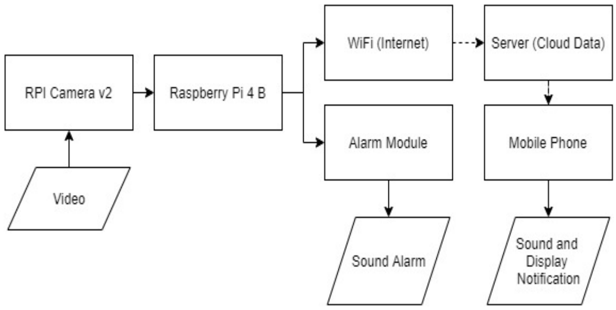


Fig. 1. Block Diagram of the System

(a) (b)

Fig. 2. Actual System (a) Hardware (b) Mobile Application

Figure 2(a) shows the actual prototype hardware that is composed of Raspberry Pi Module 4 Model B, Raspberry Pi Camera Version 2, and piezo buzzer. The prototype was placed in an elevated area high enough to cover the standard room size. This is essential to allow the system prototype to detect the target object (human) in the room and place critical points on the body. Figure 2(b) shows the home page of the mobile application designed and created with Kodular. Kodular is a platform where android applications can be created using object-oriented programming in design and code blocks with the backend.

B. MediaPipe Pose

MediaPipe Pose is an ML pipeline that predicts the high-fidelity body pose tracking from an RGB video that utilized BlazePose and MobileNetv2 CNN architecture. This is a lightweight versatile model architecture that can be easily implemented on a mobile application. BlazePose is a machine learning model that consists of a detector and an estimator developed by Google that can compute (x,y,z) coordinates of 33 skeletal points. The CNN model architecture of this model is MobileNetv2-like blocks. The evaluation of the model is relatively high performance depending on the usage, which in turn selects whether the lite/full/heavy model is the most appropriate [40]. Convolutional Neural Network (CNN), is one of the popular algorithms for deep learning. Convolution neural networks utilize the image's matrix component in order to extract data from it. Compared to other neural networks, Convolution neural networks are superior in terms of image, speech, and audio signals. As shown in Figure 3, there are 3 layers in CNN namely: Convolution Layer, Pooling Layer, and the Fully connected (FC) layer. Convolution layer is a vital part of the CNN as it is in this layer where all the major computations are executed [32]. Pooling layer is where another set of convolution layers is executed until the desired data set is captured. With each consecutive convolution layer in the pooling layer, the neural network starts to capture larger and more detailed parts of the image until the intended data is captured [33]. There are many CNN models with different architecture types and some of which are Mobile Net, Inception, and Resnet [34]. MediaPipe Pose utilizes MobileNetv2 CNN architecture.

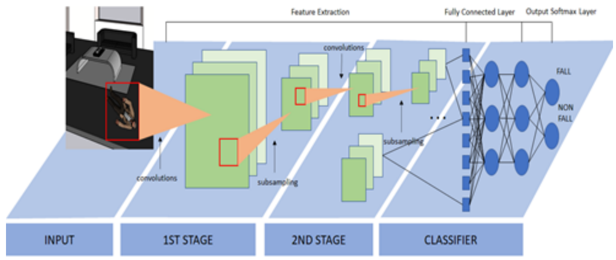
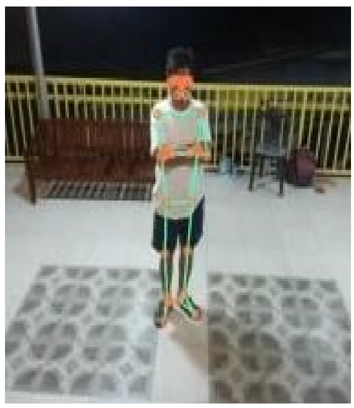


Fig. 3. Block Diagram of the Convolutional Neural Network

The image shown in Figure 4 is a sample captured frame with skeleton points. The lines and the points are plotted with colors that make up the skeleton representation. In the background, while it is plotted, it outputs a data of 1 by 33 by 4 array per frame, which represents the 33 critical points in the human body and the other four (4), which pertains to x,y,z dimension, and the v for visibility.



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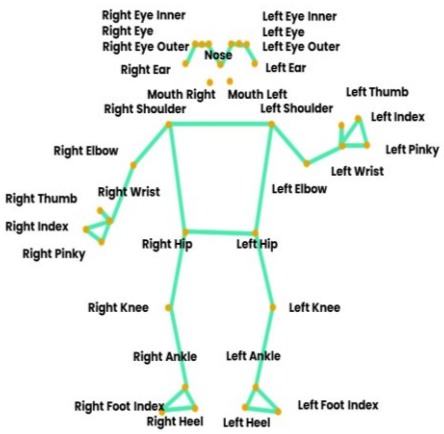


(b)

Fig. 4. Pose Points Generated on the Human Body

C. *Modeling*

1. Data capturing and labeling

The data capturing was generated through a video feed from the RPi camera v2 module. Lighting conditions, spatial resolution, and size were considered in the image acquisition. The frame rate is 30fps with a spatial resolution of 1920 by 1080 pixels. For each frame of the video data, the skeletal points using the MediaPipe Pose derived from the BlazePose model were recorded and saved to a CSV file. The human skeletal points data of male and female subjects consisting of a fall activity and non-fall activity were properly labelled and used for training the model. The researchers' adopted six poses from [28] to identify a fall action and recorded these poses from four sets of angles corresponding to the captures when the camera was placed on each of the four corners of the room. Sample images of these six poses which were performed by male and female subjects can be seen in Figure 5.

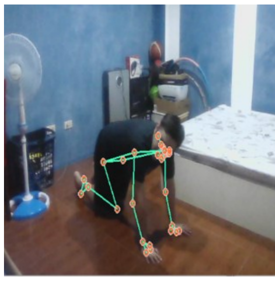
2. Training and Testing Dataset

For each frame capture, a data of 33 by 4 array was recorded which corresponds to the 33 critical points in the human body as shown in Figure 4 and the other four (4), which pertains to x,y,z vectors, and visibility. There were 17, 000 observations in total that were recorded and were divided into 70% for training and 30% for testing datasets [25], [29] for the creation of the model. The input training images were subjected to imaging techniques such as pre-processing and skeletal feature extraction to show a representation of what is expected to be detected as a fall action. The learned model was then tested by the testing datasets and the number of True Positives, True Negatives, False Positives, and False Negatives were computed and appropriate computer vision metrics [30] was used to measure the model performances such as Accuracy, F1 Score, Precision, Recall Specificity and MCC.

3. Random Forest Classifier

Using the pretrained MediaPipe Pose, the skeleton points were recorded. For any deep learning-based network, dataset selection along with proper model architecture with suitable parameters selection is very important. Using an optimized detector, the machine learning pipeline first located the person and pose region-of-interest (ROI) within the frame. The tracker subsequently predicted the pose landmarks and segmentation mask within the ROI using the ROI-cropped frame as input. The relative positions of landmarks corresponding to the 33 skeleton points were used to distinguish a fall or a non-fall pose.

The machine learning used to classify each observation is the Random Forest Classifier, which combines and uses more than 2 or more algorithms for object classification [31]. It randomly creates decision tree sets as a subset from the training set, then combines the subsets to come up with the final class for the test object. With how the RF Classifier works, gives a greater possibility of getting a more accurate and stable prediction. Even without or minimal hyper-parameter tuning, Random Forest produces great outputs due to its simplicity. The majority of new machine learning systems utilize both classification and regression, thus the RF Classifier provides the system a big advantage by including both [32]. The hyperparameters of the Random Forest classifier adopted in this current work are presented in Table 3.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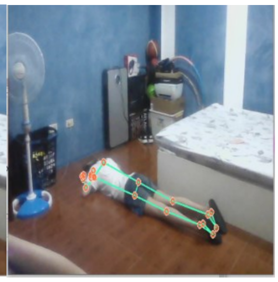
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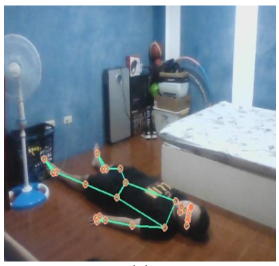
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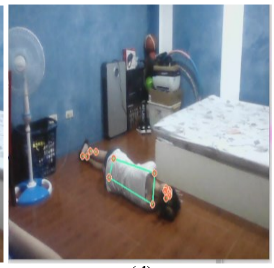
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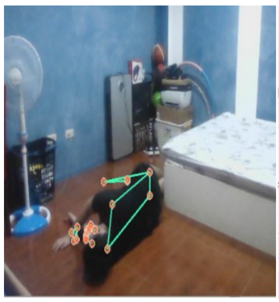
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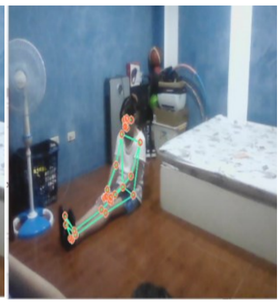
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Fig. 5. Sample Fall Poses Male and Female (a) On Four (b) Flat Forward (c) Flat Backward (d) Fall Right Side (e) Fall Left Side (f) Slightly up

TABLE 3. HYPERPARAMETERS FOR THE RANDOM FOREST CLASSIFIER

Parameter	**Value**
Number of trees	100
Quality of a split	Gini
Maximum depth of the tree	None
Minimum number of samples to split an internal node	2
Minimum number of samples required to be at a leaf node	1
Minimum weighted fraction of the sum of weights	0
Number of features for the best split	Auto
Max leaf nodes	None
Minimum impurity decrease	0
Bootstrap	True
Out-of-bag samples	False
Number of jobs to run in parallel	None
Randomness of the bootstrapping	None
Verbose	0
Reuse the solution	False
Weights associated with classes	None
Complexity parameter	0
Number of samples to draw	None

IV. RESULTS AND DISCUSSION

Experiments were performed on Raspberry Pi 4B from the videos captured using an RPi camera v2 module. Python was used as the development environment with Scikit-learn and other supplemental libraries and modules such as Gpiozero, Numpy, Requests, OS, Pandas, Pickle, CSV, Time, and Threading. MediaPipe and OpenCV were used for image processing and skeleton feature extraction. The system has been trained and tested on male and female subjects at four different camera placements. Seventeen thousand data sets with classes were divided into two categories: fall and non-fall which was split into 11,900 training data and 5,100 testing data for the creation of the model. Table 4 shows the fall detection results of the system. It can be seen that model returns a value of 4199 True Positives equal to, 0 Fales Positives, 3 False Negatives, 898 True Negatives. The developed model achieves very high performance in terms of accuracy equal to 99.94% which outperforms other vision-based methods presented in Table 2. Other metrics are also presented in Table 5 such as F1 Score of 99.96%, Precision (PPV) and Specificity (TNR) equals 100%, Recall equals 99.93%, F1 Score equals 99.96%, and MCC 99.8%.

TABLE IV. CONFUSION MATRIX

	Fall-Output	**Non-Fall Output**	
Actual Fall	TP = 4199	FN = 3	Total = 4202
Actual Non-Fall	FP = 0	TN = 898	Total = 898
	Total = 4199	Total = 901	

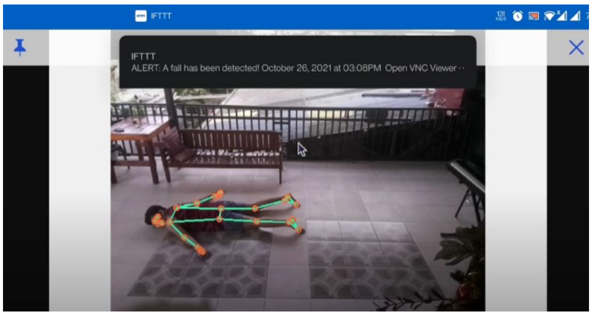
TABLE V. MODEL'S PERFORMANCE EVALUATION

Accuracy	**PPV**	**Recall**	**TNR**	**F1 Score**	**MCC**
99.94%	100%	99.93%	100%	99.96%	99.8%

The whole system functionality that includes fall detection, on-premise alarm, IoT message notification, and remote monitoring was then tested with the actual environment with an additional 80 trials considering both genders. Table 6 shows the overall performance of the system based on the 80 trials conducted.

TABLE VI. PERFORMANCE OF THE SYSTEM

System Tasks	**Accuracy**
Fall Detection	100%
On-premises Alarm	100%
IoT Message Notification & Alarm	100%
Remote Monitoring	100%



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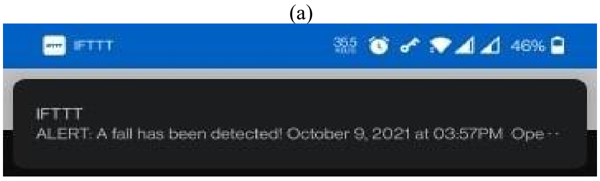
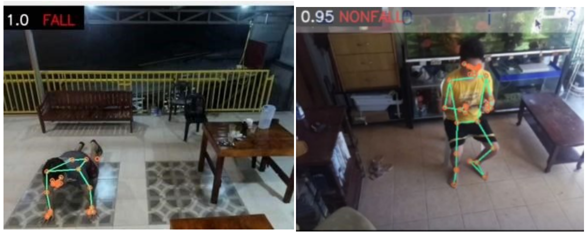


(b)

Fig. 6. Mobile Application Displays (a) sample fall as viewed on the mobile application (b) sample notification

Figure 6(a) shows the display screen of the mobile application with the current "fall" status of the human detected in an android phone. The skeleton landmark points are also shown to verify a fall action in real-time from a continuous live video feed. A sample message notification is shown in Figure 6 (b). Sample live video feed as seen on the mobile application is shown in Figure 7(a) and (b) where a box on the upper left corner indicates whether it is a fall or a non-fall activity and probability of detection.



(a) (b)

Fig. 6. Sample Live feed as viewed on the mobile application (a) fall (b) non-fall

V. CONCLUSION

This work developed a machine vision-based fall detection system using a pretrained CNN MediaPipe Pose and Random Forest Classifier with IoT monitoring and alarm. Using an embedded RPi 4B with RPi camera v2, training and testing datasets of fall and non-fall pose of both genders at four camera views were extracted using BlazePose skeleton points. The approach achieves a high accuracy of 99.94% in determining fall pose outperforming existing vision-based methods. The Internet of Things and RPi platform provided an on-premises auditory alarm, an IoT message notification, and real-time video feed via an android application. Nevertheless, extending the research can be done by investigating other machine learning algorithms on fall detection.

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